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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VII Semester B. E. Examinations March -2022

Computer Science and Engineering

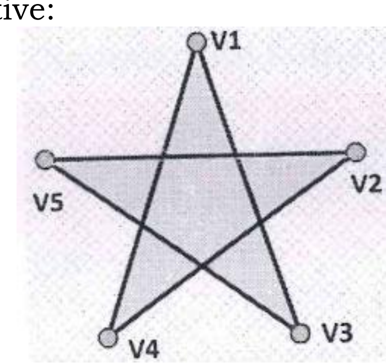
COMPUTER GRAPHICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

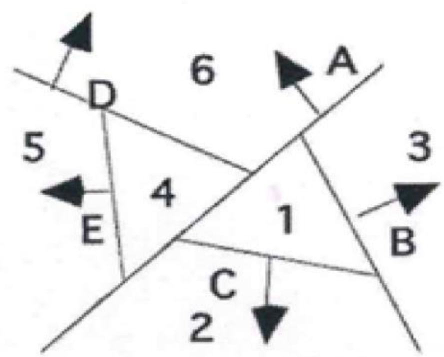
PART-A

1	1.1	Colour depth can be defined as __________ which can be displayed on a display unit.	01
	1.2	Which are the methods used for displaying characters in computer graphics?	01
	1.3	What is frame buffer?	01
	1.4	Define resolution in computer display.	01
	1.5	Consider a raster system with the resolution of 640×480. How many pixels could be accessed per second in each of these systems by a display controller that refreshes the screen at a rate of 60 frames per second?	01
	1.6	For a point to be clipped in the given clipping rectangular area, what conditions are checked?	01
	1.7	What is the need of homogenous coordinates in matrix representation?	01
	1.8	What is the intersection point of the lines $y = 2x$ and $y = -x + 6$.	01
	1.9	Explain trivial rejection in Cohen Sutherland line clipping algorithm.	02
	1.10	Write the OpenGL code snippet to draw the given Fig. 1.10 using GL_LINE_LOOP primitive:  Fig. 1.10	02
	1.11	Compare DDA Vs. Bresenham's Line drawing algorithm.	02
	1.12	Compare parallel and perspective projection.	02
	1.13	Compare object space method and image space method of visible surface detection algorithms.	02

1.14	List the properties of B spline curves.	02

PART-B

2	a	Explain the following types of polygons in OpenGL: i. GL_TRIANGLES ii. GL_QUADS iii. GL_TRIANGLE_FAN iv. GL_QUAD_STRIP	08
	b	Write a short note on flat display devices.	04
	c	Explain RGB Color model in OpenGL.	04
3	a	Analyze Midpoint circle drawing algorithm by showing the different cases of decision parameter. Give the OpenGL C code for scan converting the complete circle.	08
	b	Illustrate Sutherland-Hodgeman polygon clipping algorithm by taking an example.	04
	c	Illustrate mapping from windows to viewpoint coordinate transformation.	04
		OR	
4	a	Apply and explain Cohen Sutherland line clipping algorithm to clip a line w.r.t. a rectangular clipping boundary. Let R be the rectangular window boundary with its left bottom corner at (-10, 5) and right top corner at (12,16). Clip the following lines. i. Line (-14, 2) to (11, 7) ii. Line (-11, 5) to (13, 8) iii. Line (11, -2) to (13, 3) iv. Line (-14,7) to (-12, 10)	08
	b	Explain Bresenham's line drawing algorithm to scan convert a line whose slope m is $0 \leq m \leq +1$. How to modify this algorithm for lines whose slope $m > 1$?	04
	c	Illustrate the Boundary filling algorithm for filling a polygon, with the pseudo code for 8-connected region filling.	04
5	a	Illustrate the general combined matrix for scaling about a fixed point (xf, yf, zf), in homogenous coordinate system. The triangle $[ABC] = [(5,3,7),(4,2,8),(10,4,9)]$ is required to be magnified to four times of its original size while keeping $C(10,4,9)$fixed. What are the coordinates of the triangle after magnification?	08
	b	The triangle $[ABC] = [(2,6),(4,2),(2,4)]$ is rotated clockwise by 90° about origin and then reflected about y-axis. What are the new coordinates of the triangle?	04
	c	If we perform a rotation θ about the x-axis followed by a translation, find the composite matrix. Does the order of performing the rotation matter? Prove your answer.	04
		OR	

6	a	Summarize the transformation that rotates an object point $Q(x, y)$ by $\theta°$ about a fixed point $P(h, k)$. Derive the general form of the composite transformation matrix for rotation about a fixed point. Write a C function (without OpenGL libraries) to perform a 45° rotation of a triangle $A(0,0)$, $B(1,1) and\ C(5,2)$ about the point $(-1, -1)$.	08
	b	Illustrate Shear transformation matrix for 3-dimensional object.	04
	c	Reflect the triangle $[ABC][(2,3),(5,8),(7,2)]$ about i. The horizontal line $y = 2$ ii. The vertical line $x = 2$ iii. The line $y = x + 2$ iv. Draw the reflected image in all the cases.	04
7	a	Write and explain Classification of planar geometric projections. Discuss the OpenGL functions supported for setting view volume in perspective and parallel projections.	08
	b	Identify the principal vanishing points for the standard perspective projection; give explanation with diagram and real world examples.	04
	c	With a neat diagram explain 3D viewing pipeline.	04
8	a	Derive an expression for a Hermite spline as a function of its control points and normal. Discuss the differences between Hermite splines, Bezier splines, and B-splines.	08
	b	Draw a BSP tree that represents the space partitioning given in Fig. 8b. Hand simulate and explain the processes of detecting the visible surfaces.  Spatial partitioning Fig. 8b.	04
	c	Explain the following hidden surface removal algorithm: i. Scan Line algorithm ii. Z buffer algorithm	04